

Unit 06 Practice Problems: Normal Distribution & CTL

Instructor: Dr. Md Mutasim Billah

Required Practice Problems: Normal Distribution

Complete these required practice problems throughout the unit as we cover the corresponding topics in class. These problems reinforce the concepts discussed in the lecture notes, handouts, and course resources. Before attempting the homework assignments and/or preparing for exams, make sure you have completed and reviewed all practice problems from the unit. The **solutions** are posted on Canvas (Unit-PP-05-06-Solution.pdf > Practice Problem-Solution > STAT 312 > Files).

5.11

Write the pdf of $N(\mu = 10, \sigma^2 = 4)$.

5.12

Identify the values of μ and σ^2 of a normal random variable with the following pdf:

$$f(x) = \frac{1}{9\sqrt{2\pi}} e^{-\frac{(x+5)^2}{162}}, \quad -\infty < x < \infty.$$

5.13

X = weight of a randomly selected 12-year-old American boy.

$$X \sim N(\mu = 40 \text{ kg}, \sigma = 2.1 \text{ kg}).$$

- Use the Empirical Rule to find an interval of weights where approximately 68% of the boys have a weight in the interval.
- Use the Empirical Rule to find an interval of weights where approximately 95% of the boys have a weight in the interval.
- Use the Empirical Rule to find an interval of weights where approximately 99.7% of the boys have a weight in the interval.

5.14

$$Z \sim N(\mu = 0, \sigma = 1).$$

Draw pictures and find each of the following probabilities:

- $P(Z < -0.68)$
- $P(Z > -1.35)$
- $P(-1.24 < Z < 0.87)$
- $P(Z < 1.79)$
- $P(Z > 0.37)$
- $P(-2.05 < Z < 0)$

5.15

$$Z \sim N(\mu = 0, \sigma = 1).$$

Draw pictures and find each of the following values:

- (a) Find the value of a if $P(Z < a) = 0.9906$.
- (b) Find the value of b if $P(Z > b) = 0.0708$.
- (c) Find the value of c if $P(-c < Z < c) = 0.3830$.
- (d) Find the value of d if $P(d < Z < 0.82) = 0.7271$.
- (e) Find the 48th percentile on the Z curve. What is the 52nd percentile?
- (f) Find the 89th percentile on the Z curve. What is the 11th percentile?

5.16

X = weight of a randomly selected 12-year-old American boy.

$$X \sim N(\mu = 40 \text{ kg}, \sigma = 2.1 \text{ kg}).$$

Draw pictures.

- (a) $P(X < 38.53) = ?$
- (b) $P(38.32 < X < 41.89) = ?$
- (c) What is the 91st percentile among the 12-year-old boys' weights?

5.17

Engineers have designed a process to fill plastic milk jugs.

X = amount of milk dispensed into a plastic jug (gallons).

$$X \sim N(\mu = 1.02, \sigma = 0.01).$$

Draw pictures.

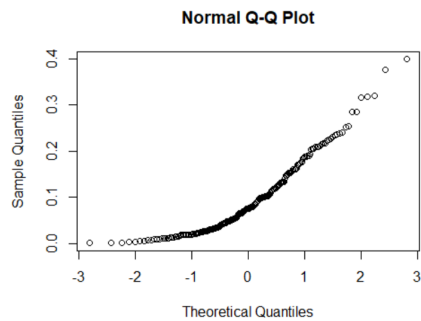
- (a) If a milk jug is considered underfilled if it contains less than 1 gallon of milk, what proportion of milk jugs will be underfilled?
- (b) Milk jugs should not be overfilled because that would reduce profitability and possibly threaten the capacity of the milk jugs themselves. If a milk jug contains more than 1.0436 gallons, it is considered overfilled. What proportion of milk jugs will be overfilled?
- (c) 61% of the milk jugs will contain more than k gallons of milk. What is the value of k ?

5.18

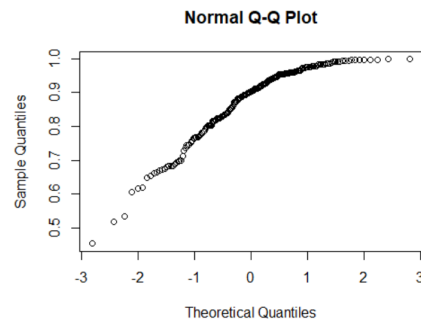
Three QQ plots were generated using three different data sets. Choose exactly one of the following descriptions for each data set:

- The data set appears to be normally distributed.
- The data set appears to be heavy tailed.
- The data set appears to be weak tailed.
- The data set appears to be right-skewed.
- The data set appears to be left-skewed.

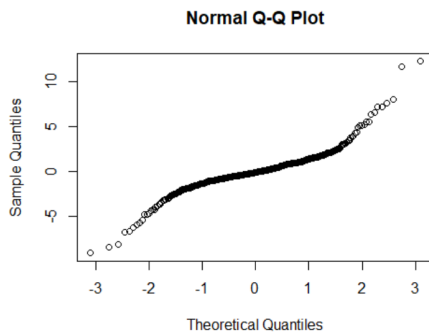
(a)



(b)



(c)



Required Practice Problems: Central Limit Theorem

Complete these required practice problems throughout the unit as we cover the corresponding topics in class. These problems reinforce the concepts discussed in the lecture notes, handouts, and course resources. Before attempting the homework assignments and/or preparing for exams, make sure you have completed and reviewed all practice problems from the unit. The **solutions** are posted on Canvas ([Unit-PP-06-Solution.pdf](#) > [Practice Problem-Solution](#) > [STAT 312](#) > [Files](#)).

6.1

Refer to Practice Problem 5.16.

X = weight of a randomly selected 12-year-old American boy.

$$X \sim N(\mu = 40 \text{ kg}, \sigma = 2.1 \text{ kg})$$

Instead of randomly selecting one boy, suppose more than one boy is randomly selected and the sample mean weight among them is calculated.

(a) If $n = 4$, find $P(\bar{X} > 41)$. Draw a picture.

(b) If $n = 8$, find $P(\bar{X} > 41)$. Draw a picture.

6.2

Refer to Practice Problem 5.17.

Engineers have designed a process to fill plastic milk jugs.

X = amount of milk dispensed into a plastic jug (gallons).

$$X \sim N(\mu = 1.02, \sigma = 0.01)$$

Instead of randomly selecting one milk jug, suppose more than one milk jug is randomly selected and the sample mean among them is calculated.

(a) What is the probability the sample mean amount of milk among $n = 7$ randomly selected milk jugs will be less than 1.01 gallons? Draw a picture.

(b) What is the probability the sample mean amount of milk among $n = 3$ randomly selected milk jugs will be between 1.015 and 1.022 gallons? Draw a picture.

6.3

Refer to Practice Problem 5.7 (from Unit 05 PP).

X = circumference of a randomly selected baseball (inches).

$$X \sim \text{UNIF}(9.0, 9.1)$$

Instead of randomly selecting one baseball, suppose more than one baseball is randomly selected and the sample mean circumference among them is calculated.

(a) If $n = 30$, describe the approximate distribution of the sample mean.

(b) If $n = 30$, approximate $P(9.04 < \bar{X} < 9.06)$. Draw a picture.

(c) If you were to continue approximating $P(9.04 < \bar{X} < 9.06)$ with larger and larger sample sizes, do you think the probability would increase or decrease?